

Exploratory Data Analysis

PYTHON FOR DATA ANALYSIS

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1. Display the records, first 10 records, last 10 records.

```
import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
print(df)
df.head(10)
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	DrugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	DrugY
..	...	..	...	...	...	...
195	56	F	LOW	HIGH	11.567	drugC
196	16	M	LOW	HIGH	12.006	drugC
197	52	M	NORMAL	HIGH	9.894	drugX
198	23	M	NORMAL	NORMAL	14.020	drugX
199	40	F	LOW	NORMAL	11.349	drugX

[200 rows x 6 columns]

[9]:

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	DrugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	DrugY
5	22	F	NORMAL	HIGH	8.607	drugX
6	49	F	NORMAL	HIGH	16.275	DrugY
7	41	M	LOW	HIGH	11.037	drugC
8	60	M	NORMAL	HIGH	15.171	DrugY
9	43	M	LOW	NORMAL	19.368	DrugY

```
[10]: import pandas as pd
df = pd.read_csv(r"C:\Users\siraj\Downloads\drug200.csv")
print(df)
df.head(10)
df.tail(10)
```

```
[10]:
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
190	58	M	HIGH	HIGH	18.991	DrugY
191	23	M	HIGH	HIGH	8.011	drugA
192	72	M	LOW	HIGH	16.310	DrugY
193	72	M	LOW	HIGH	6.769	drugC
194	46	F	HIGH	NaN	34.686	DrugY
195	56	F	LOW	HIGH	11.567	drugC
196	16	M	LOW	HIGH	12.006	drugC
197	52	M	NORMAL	HIGH	9.894	drugX
198	23	M	NORMAL	NORMAL	14.020	drugX
199	40	F	LOW	NORMAL	11.349	drugX

2. Column Data Types

```
[13]: import pandas as pd
df = pd.read_csv(r"C:\Users\siraj\Downloads\drug200.csv")
df['Age'].dtype
```

```
[13]: dtype('int64')
```

```
[14]: import pandas as pd
df = pd.read_csv(r"C:\Users\siraj\Downloads\drug200.csv")
df['Age'].dtype
df.dtypes
```

```
[14]: Age          int64
Sex            str
BP             str
Cholesterol     str
Na_to_K        float64
Drug           str
dtype: object
```

3. Find the number of records

```
[15]: import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
df.shape
print("Total records:", len(df))
```

Total records: 200

4. How many elements are there?

```
[16]: import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
df.size
```

[16]: 1200

5. Provide statistical summary

```
[17]: import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
df.describe()
```

```
[17]:
```

	Age	Na_to_K
count	200.000000	200.000000
mean	44.315000	16.084485
std	16.544315	7.223956
min	15.000000	6.269000
25%	31.000000	10.445500
50%	45.000000	13.936500
75%	58.000000	19.380000
max	74.000000	38.247000

6. List all columns, data types and check missing values

```
[18]: import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Age         200 non-null    int64
1   Sex         200 non-null    str
2   BP          200 non-null    str
3   Cholesterol  199 non-null    str
4   Na_to_K     200 non-null    float64
5   Drug        199 non-null    str
dtypes: float64(1), int64(1), str(4)
memory usage: 9.5 KB
```

7. Viewing specific rows and columns

```
[19]: import pandas as pd
df = pd.read_csv(r"C:\Users\suraj\Downloads\drug200.csv")
print(df.iloc[0])
print(df.iloc[5:10])
```

```
Age          23
Sex          F
BP          HIGH
Cholesterol  HIGH
Na_to_K      25.355
Drug        DrugY
Name: 0, dtype: object
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
5	22	F	NORMAL	HIGH	8.607	drugX
6	49	F	NORMAL	HIGH	16.275	DrugY
7	41	M	LOW	HIGH	11.037	drugC
8	60	M	NORMAL	HIGH	15.171	DrugY
9	43	M	LOW	NORMAL	19.368	DrugY